



Speech Anatomy 1

Imaging

Dr. Josef Fruehwald

2026-08-27

Seeing Speech

Maybe you think speech looks like this:

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ɸ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Seeing speech

Speech is a physical activity

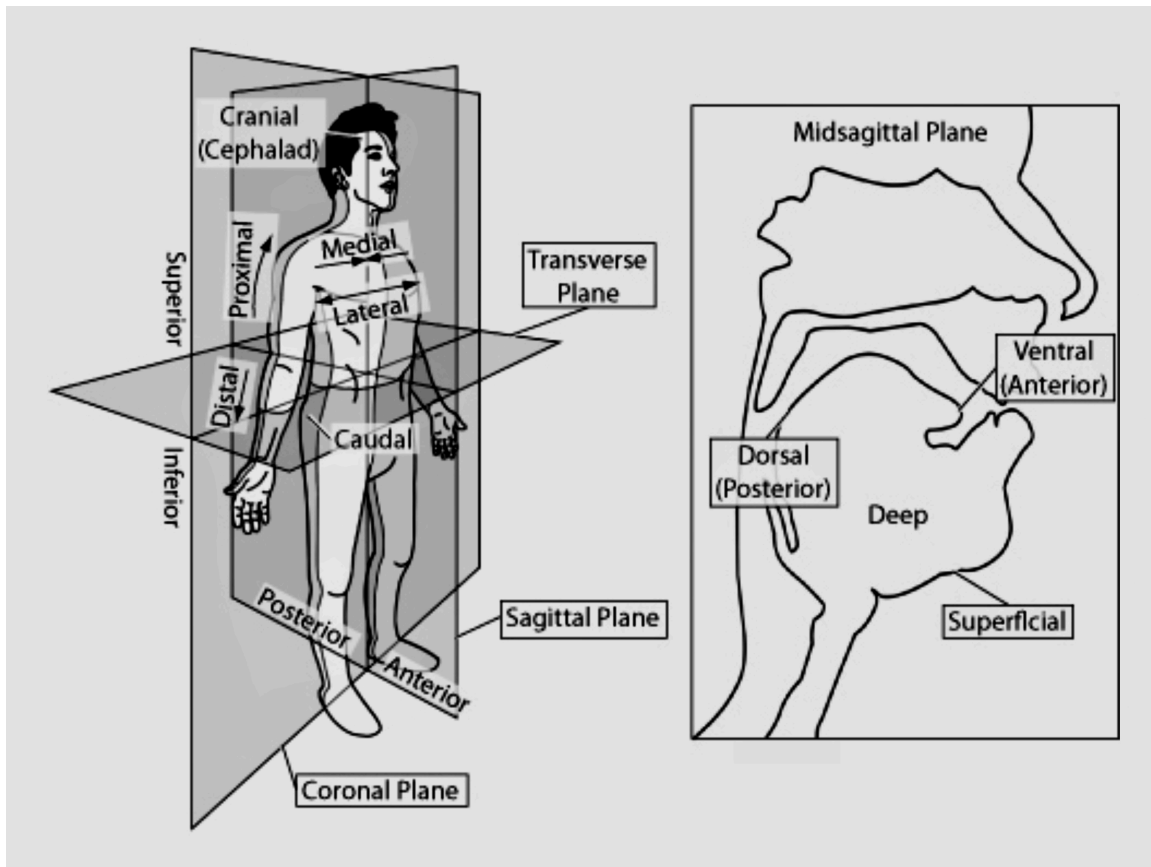
videos/xray.webm

Today

- Tools we use for looking at speech.
- Starting some anatomy.

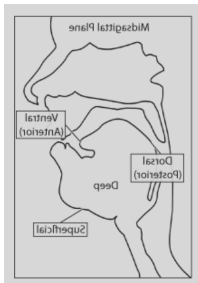


Conceptualizing the head



Posterior Anterior
 “back” “front”

Conceptualizing the head



	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill				r					ʀ		
Tap or Flap		v		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant	w	ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

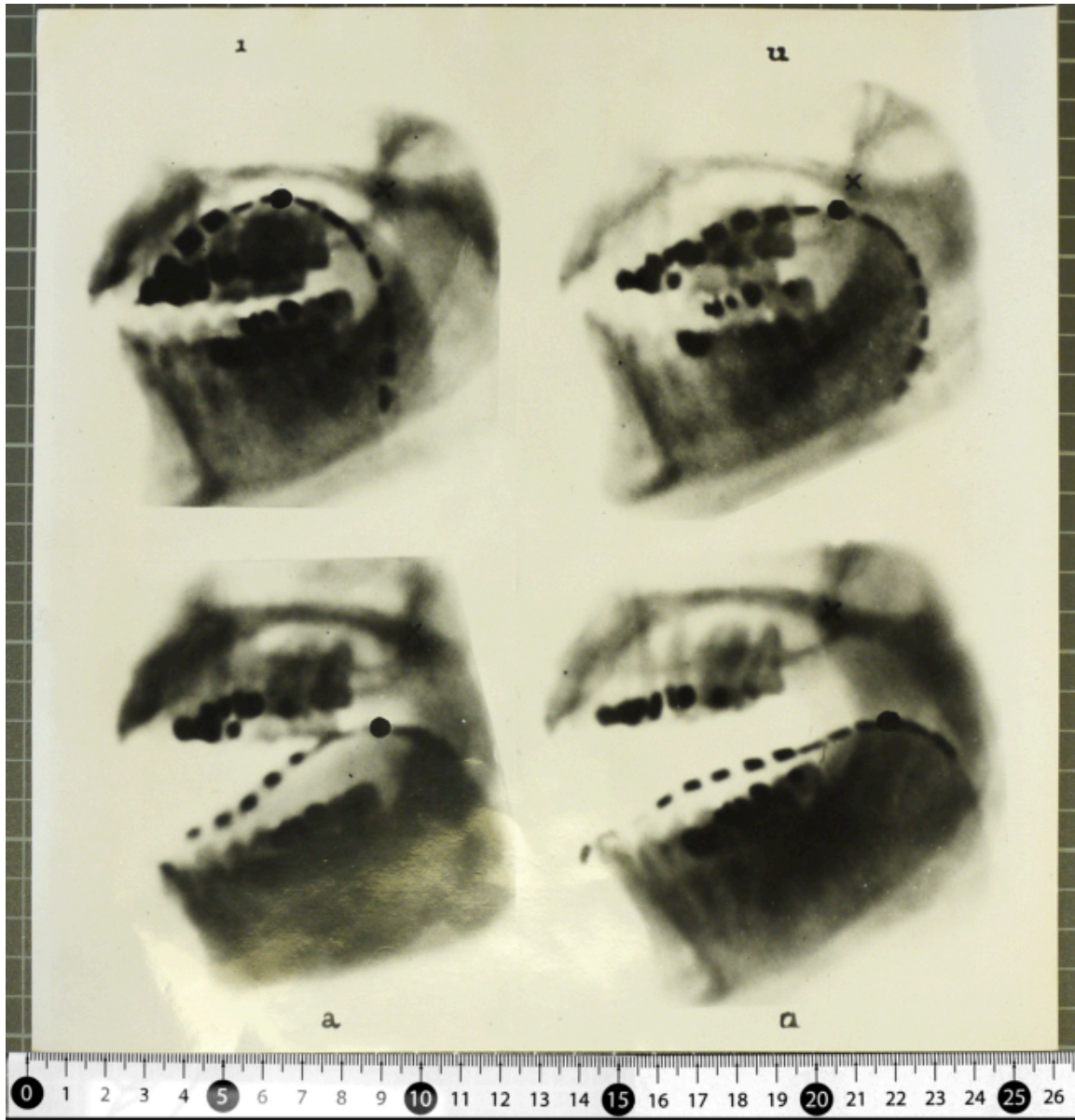


Tools of the trade

X-Ray

anterior ←

→ posterior





Tools of the trade

X-Ray

Pros?

- Better than nothing

Cons?

- Can't image fleshy bits well
- Harmful to health
- Static/low framerate

Tools of the trade

Ultrasound tongue imaging

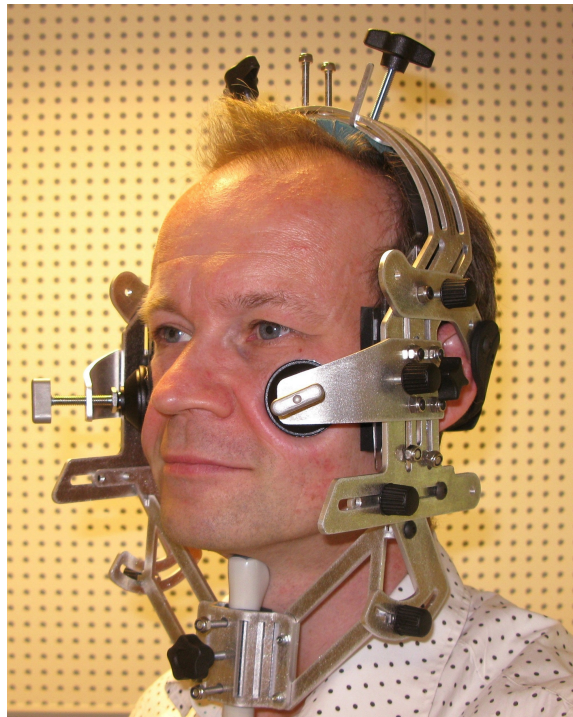


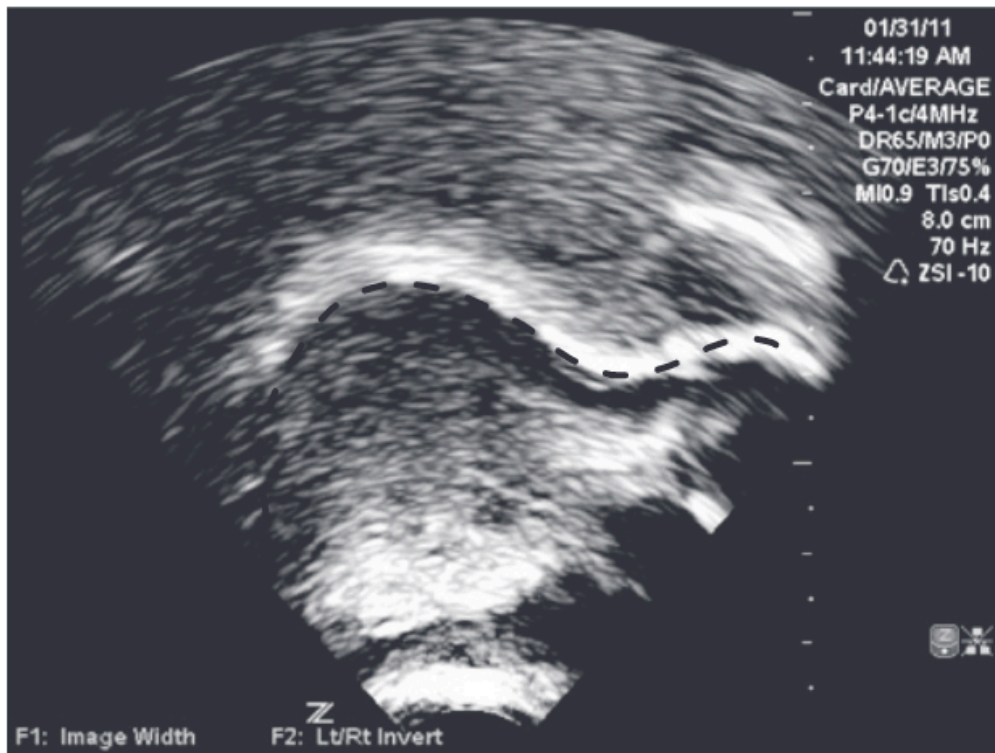
Figure 1: source: <https://www.articulateinstruments.com/ultrasound-imaging/?target=Micro%2FSonoSpeech>

Tools of the trade

Ultrasound tongue imaging

posterior ←

→ anterior



Lin et al. (2014)

Tools of the trade

Ultrasound tongue imaging

Pros?

- Relatively non-invasive.

Cons?

- Can *only* image fleshy bits.
- Can't image across an air gap.
- Relatively low frame rate.



Tools of the trade

Electromagnetic Articulography (EMA)

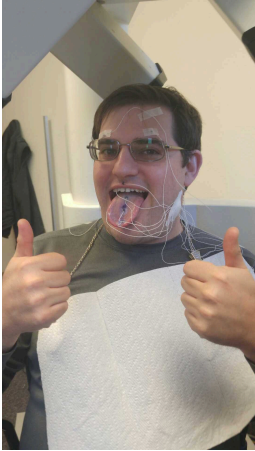
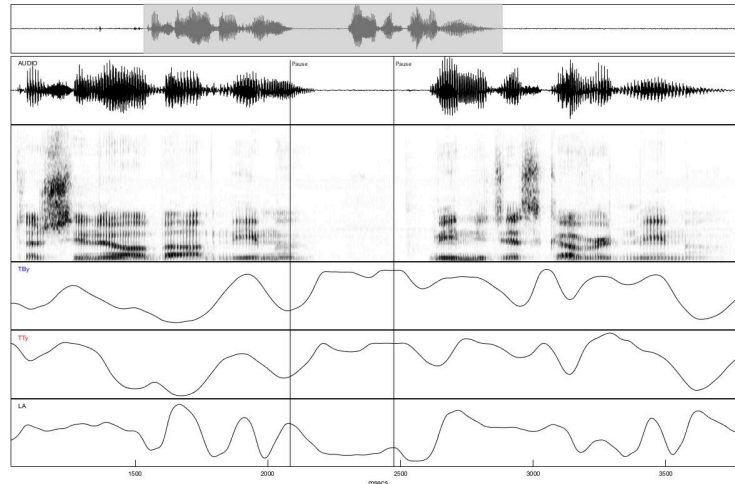


Figure 1: Credit:
Will Styler



Tools of the trade

Electromagnetic Articulography (EMA)

Pros?

- Very high time resolution.
- Spatially precise.

Cons?

- Invasive.
- Harder to visualize.
- Expensive equipment.

Tools of the trade

MRI

videos/mri.mp4

posterior

anterior



Tools of the trade

MRI

Pros?

- Non invasive.
- High quality images.

Cons?

- Low frame rate (but improving).
- Very loud.
- Speakers must be lying down.
- Expensive equipment.

Seeing Speech

The International Phonetic Alphabet (revised to 2005)

There are two MRI viewing options for both charts: MRI 1 and MRI 2.

- **MRI 1** was recorded in 2014 at Edinburgh Imaging Facility Queen's Medical Research Institute (QMRI), with a frame rate of 7.5fps. Each file has been overdubbed with "clean" audio, recorded in a recording studio.
- **MRI 2** was recorded in 2022-2023 at the Edinburgh Imaging Facility, Royal Infirmary Edinburgh (RIE), with a frame rate of 6.25fps and a 70% increase in image resolution, compared to MRI 1. Audio for the MRI 2 files was recorded in the MRI machine using an optical microphone and post-hoc noise cancelling. We aim to replace the MRI 2 audio with clean audio in the near future.

Consonants (Pulmonic) Consonants (Non-Pulmonic) Vowels Other Symbols

Consonants (Pulmonic - produced with air from the lungs)

Video type: [Animation](#) [MRI 1](#) [MRI 2](#) [Ultrasound](#) Speaker: [Janet Beck](#)

Press on a symbol to play the associated video.

	Bilabial		Labiodental		Dental		Alveolar		Postalveolar		Retroflex		Palatal		Velar		Uvular		Pharyngeal		Glottal	
Plosive	p	b					t	d			ʈ	ɖ	c	ɟ	k	g	q	ɢ		ʕ		
Nasal		m		ɱ			n				ɳ		ɲ		ŋ		ɴ					
Fricative							r										ʁ					
Tap or Flap				v			ɾ					ɽ										
Fricative	ɸ	β	f	v	θ	ð	s	z	ʃ	ʒ	ʂ	ʐ	ç	ʝ	x	χ	ʁ	ħ	ʕ	h	ɦ	
Lateral fricative							ɬ	ɮ														
Approximant				ʋ			ɹ					ɻ		j		ɰ						
Lateral approximant							l					ɭ		ʎ		ʟ						

References

Bibliography

Lin, S., Beddor, P. S., & Coetzee, A. W. (2014). *Gestural reduction , lexical frequency , and sound change : A study of post-vocalic / l /*. 5(1), 9–36. <https://doi.org/10.1515/lp-2014-0002>